

AYON ROY

PERSONAL INFORMATION

Permanent Address	Chandpur, Bangladesh 3600
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RESEARCH PROFILE

Profile	M1 student in Cyber-Physical Systems and Services with research experience in efficient, reliable, and adaptive AI. My work spans architecture-aware neural-network pruning, structured sparsity optimisation, scalable synthetic-data pipelines, interpretable learning for mobile-network operations, multilingual machine translation, and zero-retraining guardrail agents. I am interested in developing efficient, trustworthy, and human-centred AI for cyber-physical systems, including IoT and edge intelligence, intelligent sensing, digital twins, autonomous systems, and secure connected environments.
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EDUCATION

09/2026–Present	Master of Science (M1) in Cyber-Physical Systems and Services (CPS2) <i>Université Jean Monnet, Saint-Étienne, France</i>
10/2020–02/2025	Bachelor of Science in Computer Science and Engineering <i>BRAC University, Dhaka, Bangladesh</i> 136 credits (four years) • CGPA: 3.90/4.00 (Highest Distinction)

ENGLISH PROFICIENCY TEST

IELTS	Band 7.5 / CEFR Level C1 Listening: 8.0 • Reading: 7.5 • Writing: 6.5 • Speaking: 7.0
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THESIS INFORMATION

Advancing Bangla Machine Translation Through Informal Datasets

Available at: doi.org/10.48550/arXiv.2512.13487

Created an informal Bangla–English sentence dataset and trained hybrid and deep-learning models to improve translation quality for low-resource and informal language. The dataset was manually created and translated from social-media conversations and made publicly available for further research.

RESEARCH

Adaptive Alarm Threshold Prediction in 4G Mobile Networks: A Percentile-Guided Deep Learning Framework with Interpretable Outputs

Available at: doi.org/10.48550/arXiv.2605.00838 (Under peer review at the *Journal of Network and Computer Applications*)

Presents a deep-learning approach to predicting suitable alarm thresholds for 4G mobile networks based on network behaviour. The model makes alarm settings more adaptive and accurate while using fewer parameters than existing models.

Beyond Pandas: A Dask-Based Parallel Pipeline for Large-Scale Synthetic Data Generation

Accepted at the *ICCA conference*; forthcoming in the *ACM Digital Library*.

Presents a Dask-based parallel approach for generating large volumes of synthetic financial data more efficiently than Pandas, reducing execution time and memory usage while maintaining statistical quality.

Layerwise Sparsity Allocation in Convolutional Neural Networks via a Linear-Chain CRF Energy Model

Accepted at the *ICCA conference*; forthcoming in the *ACM Digital Library*.

Proposes a CNN-pruning method that determines how much to remove from each layer while accounting for dependencies between neighbouring layers. The Chain-CRF method improves performance at high compression levels while preserving model accuracy.

Probe-Reliability and Architectural Compensation in CNN Channel Pruning

Accepted at the *MIET conference*; forthcoming in a *Scopus-indexed Springer Nature book*.

Investigated reliable methods for identifying removable CNN channels while preserving model performance, demonstrating that channel-importance measures are architecture-dependent and proposing Group-Taylor scoring to improve pruning decisions in depthwise and SE-gated networks.

RELEVANT PROJECTS

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| Academic | <ul style="list-style-type: none">● Online Payments Fraud Detection: Classified fraudulent online transactions using K-nearest neighbours, logistic regression, and naive Bayes.● Detecting Sarcasm in English: Classified slang and sarcasm in English sentences using GRU and LSTM models.● Blind Stick with Morse-Code Response and GSM: Built an assistive smart-stick prototype using Arduino and multiple sensors. |
| Industrial | <ul style="list-style-type: none">● 2G Dimensioning Busy-Hour Traffic Prediction: Developed a SARIMA-based system with a front end that uses cleaned historical data and filters to forecast traffic, improving service quality and reducing call drops by approximately 75%.● Fluctuating Cell Detection and Responsible-Team Notification: Built a Python-based automated system that detects fluctuating network cells every six hours, flags issues based on defined thresholds, and emails the responsible teams, enabling faster responses and reducing operational costs. |

EXPERIENCE

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| 06/2025–06/2026 | Adjunct Lecturer
<i>BRAC University, Dhaka, Bangladesh</i>
Designed examinations, assessments, and course content, and delivered lectures on Object-Oriented Programming, Algorithms, Artificial Intelligence, and Introduction to Robotics.
Faculty profile: cse.sds.bracu.ac.bd/faculty_profile/436/ayon_roy |
| 05/2025–08/2025 | Digitalyst Intern
<i>AI and Automation Department, Banglalink Digital Communications Ltd, Dhaka, Bangladesh</i>
Developed automation tools, including a Python-based cell-monitoring and alert system, a 2G half-rate dimensioning prediction model using time-series data with a web interface, and scripts integrating Google Earth data into the Customer Complaint Automation back end. Created internal documentation, including GitLab guides for employees. |
| 03/2025–04/2025 | Global Intern
<i>Excelebrate, Remote</i>
Collaborated with international students on data cleaning using Python and SQL, exploratory data analysis (EDA), ETA reporting, the construction of master tables in PostgreSQL, and data visualisation using tools such as Looker Studio. |
| 03/2025–06/2025 | Academic Intern
<i>Information and Communication Technology Division, NCC Bank PLC, Dhaka, Bangladesh</i>
Worked with the machine-learning team to perform systematic data preprocessing and tune the parameters of a deep-learning-based fraud-detection model for mobile financial service fund transfers. |
| 06/2023–09/2024 | Undergraduate Student Tutor / Teaching Assistant
<i>Department of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh</i>
Supported students in programming fundamentals, Discrete Mathematics, and logical problem-solving. Collaborated with faculty to assess assignments, review quizzes, and manage grade records. |

SKILLS

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| Research & teaching | Research methods, academic writing, student engagement, assessment, and feedback |
| AI & data | Natural language processing, large language models, algorithms, agents, image processing, and machine learning for time-series, text, and image data |
| Security | Cryptographic models and encryption and decryption techniques |
| Professional | Teamwork and leadership in multicultural industries and international settings |

EXTRA-CURRICULAR ACTIVITIES

- Effective Leadership certification by HP LIFE
- Member, BRAC University Computer Club
- Volunteer, BYSO (Bangladesh Youngster Social Organization)

REFERENCES

Dr. Pierre Maret
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University Jean Monnet
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